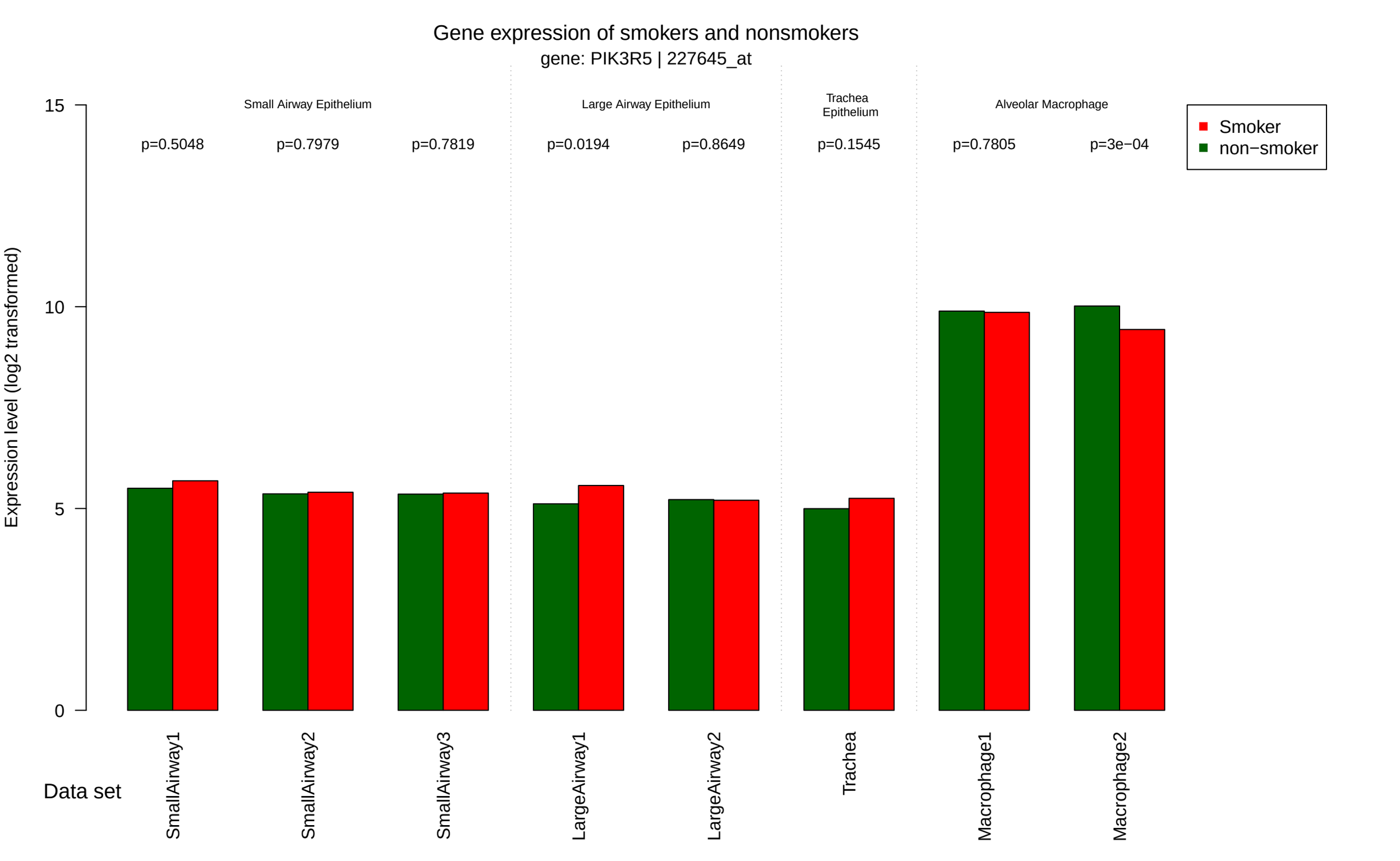
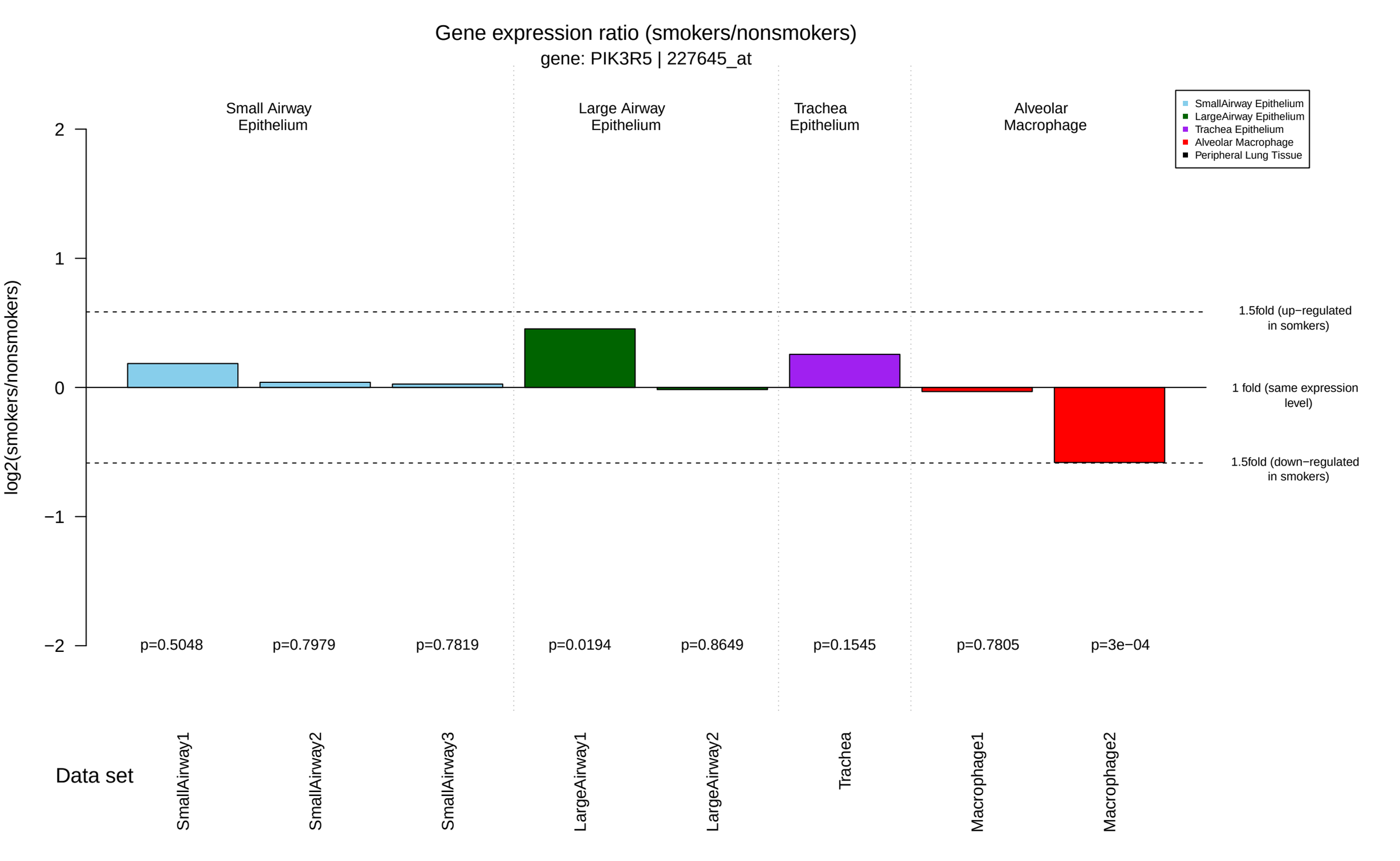
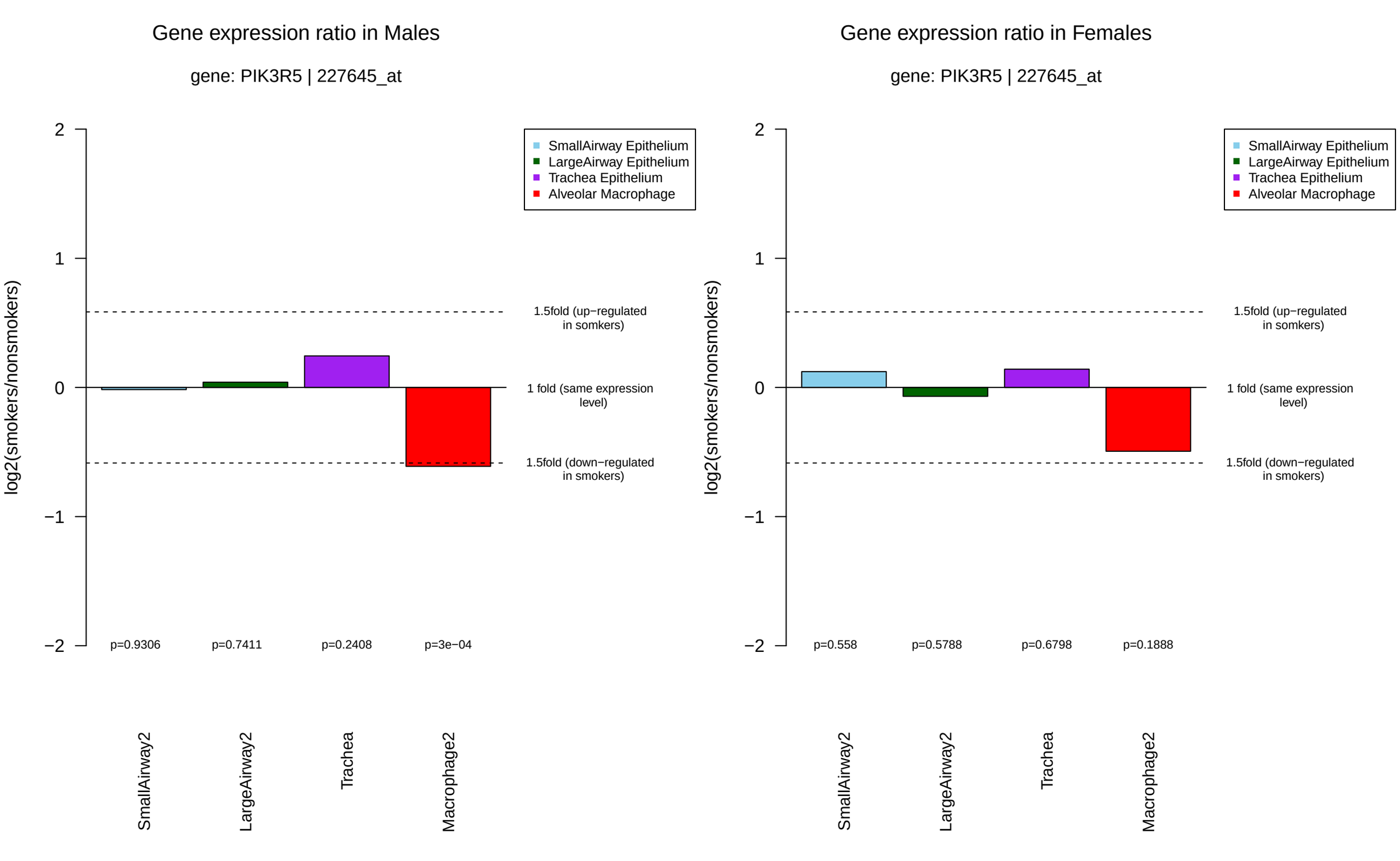
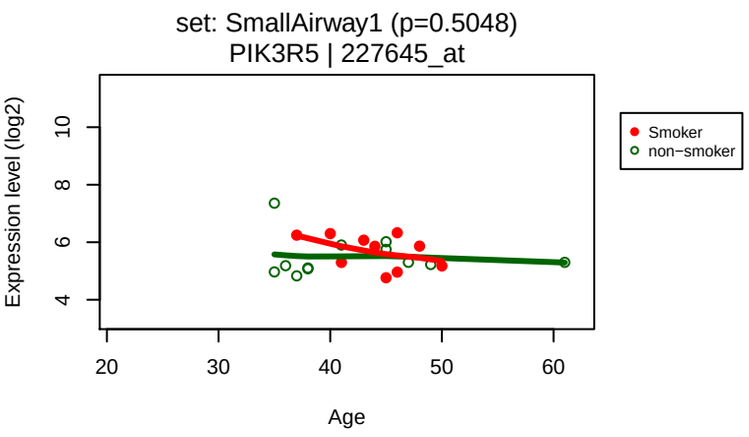
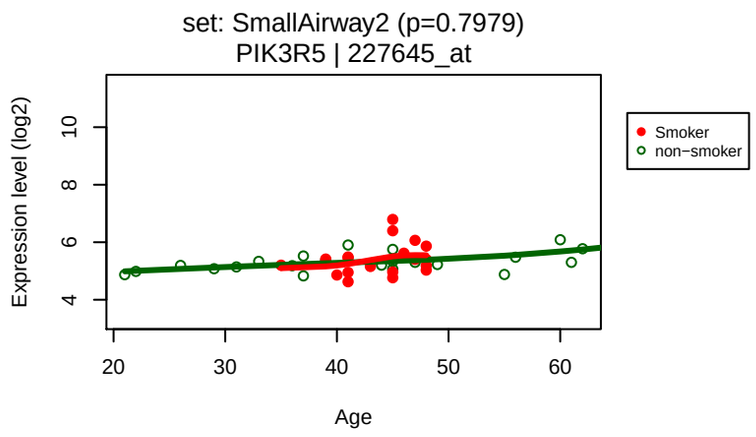
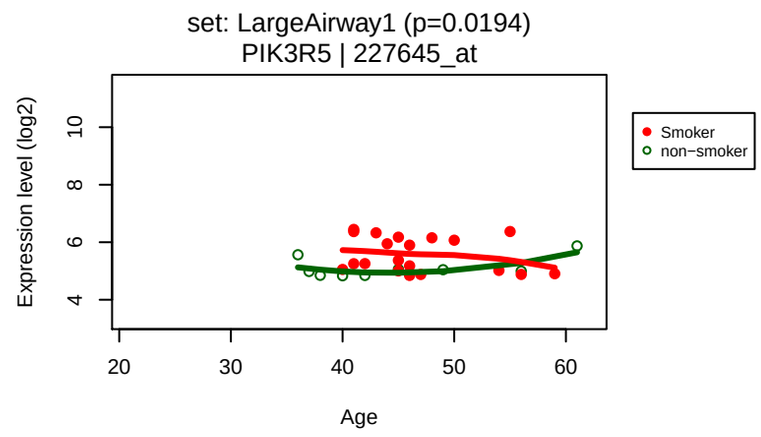

Gene expression of smokers and nonsmokers
gene: PIK3R5 | 227645_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5048
p=0.7979
p=0.7819
p=0.0194
p=0.8649
p=0.1545
p=0.7805
p=3e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

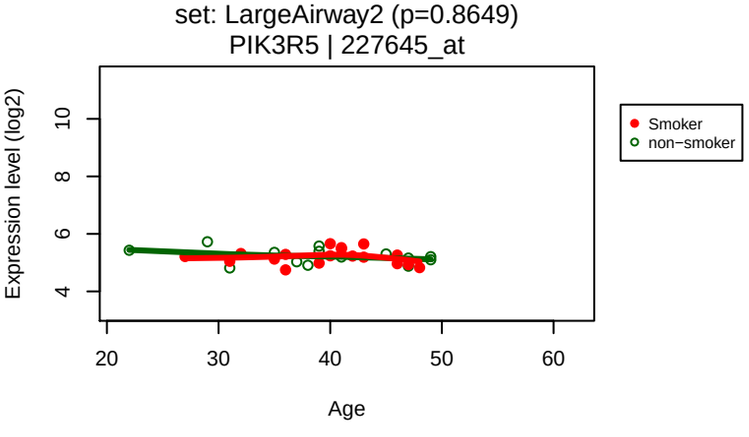

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R5 | 227645_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5048
p=0.7979
p=0.7819
p=0.0194
p=0.8649
p=0.1545
p=0.7805
p=3e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

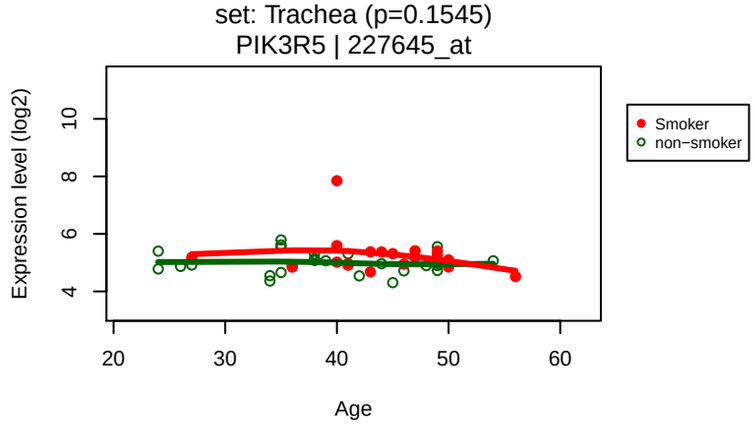

Gene expression ratio in Males
gene: PIK3R5 | 227645_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9306
p=0.7411
p=0.2408
p=3e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R5 | 227645_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.558
p=0.5788
p=0.6798
p=0.1888
SmallAirway2
LargeAirway2
Trachea
Macrophage2

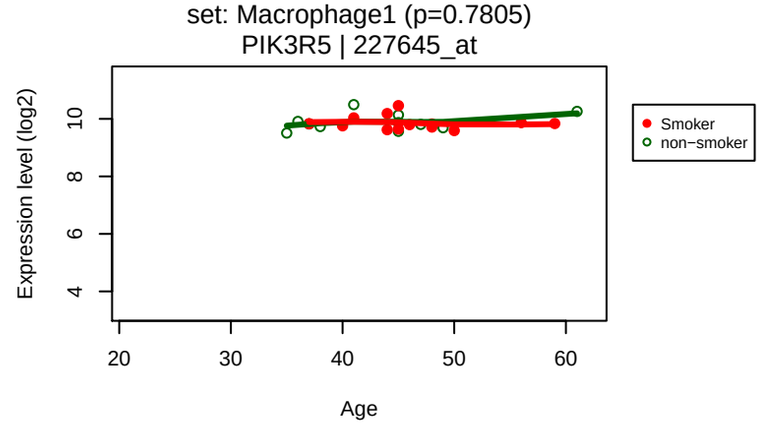

set: SmallAirway1 (p=0.5048)
PIK3R5 | 227645_at
Expression level (log2)
Age
Smoker
non−smoker

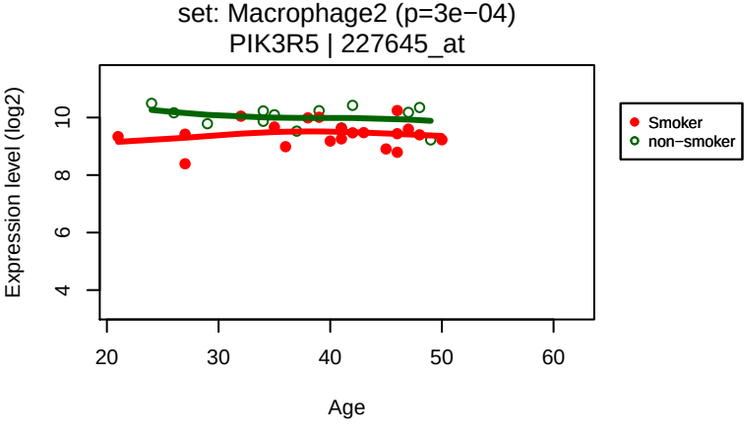

set: SmallAirway2 (p=0.7979)
PIK3R5 | 227645_at
Expression level (log2)
Age
Smoker
non−smoker

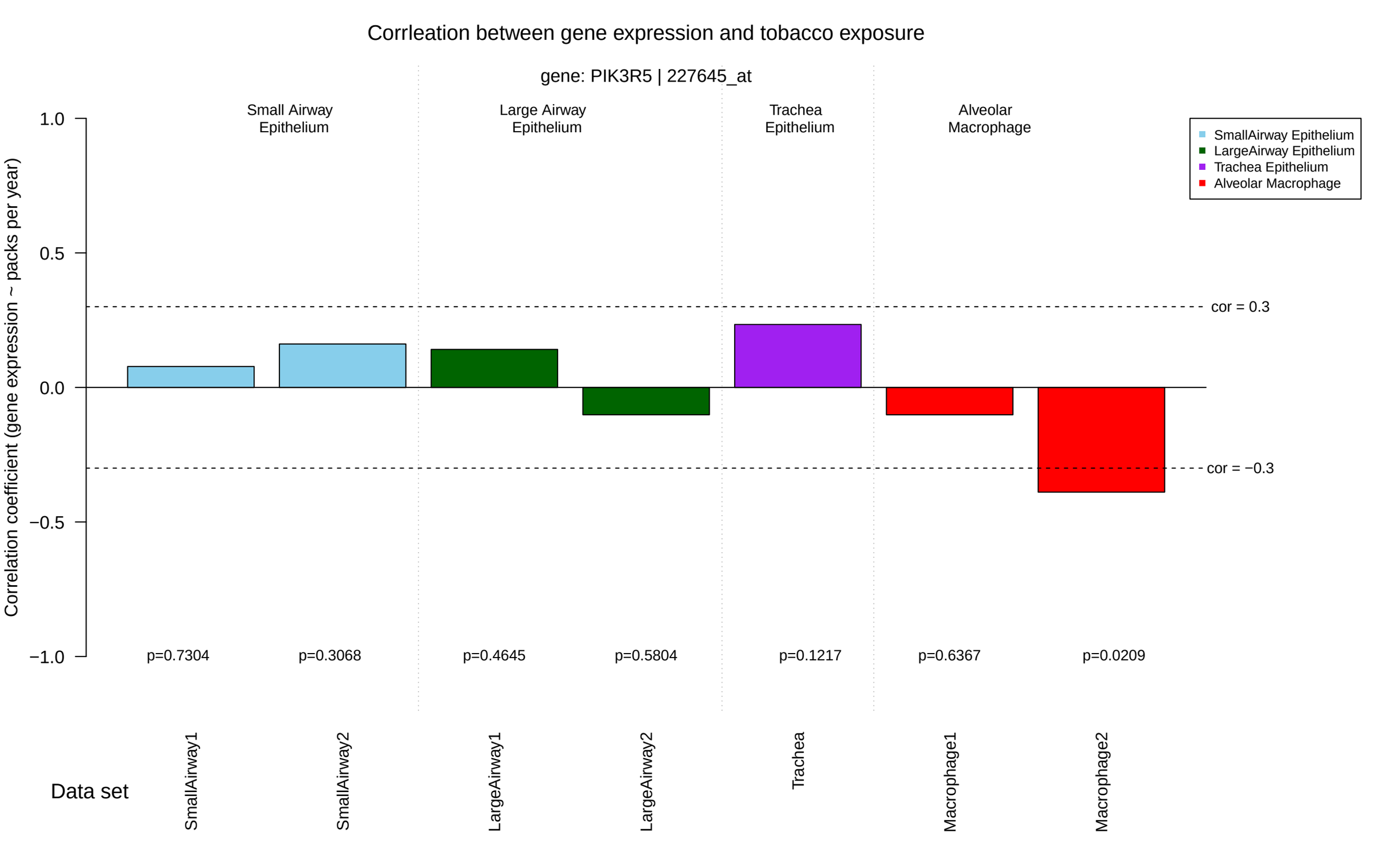

set: LargeAirway1 (p=0.0194)
PIK3R5 | 227645_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8649)
PIK3R5 | 227645_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1545)
PIK3R5 | 227645_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7805)
PIK3R5 | 227645_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=3e−04)
PIK3R5 | 227645_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R5 | 227645_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7304
p=0.3068
p=0.4645
p=0.5804
p=0.1217
p=0.6367
p=0.0209
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2